Dong 2020 Prostate — analysis summary

GO / NO-GO GATES

Gate 1 — mixing & continuity → GO

Samples intermix per-neighbourhood (iLISI 1.93 of 4 at θ=10, up from 1.0 pre-Harmony) on ONE connected component (kNN + independent UMAP radius graph); 9,698 malignant cells, no islands removed.

Gate 2 — module geography → GO (non-branching)

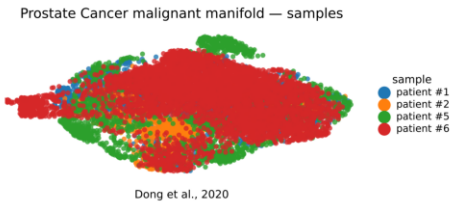
Shared Gast/pSom gradient (smoothed r=+0.90, co-vary) with Crest an inverse terminal readout (r=−0.39 / −0.53). Curve fold-free (0 forks); Crest-high decile at pseudotime 0.98.

Gate 3 — hub cross-repression → NA

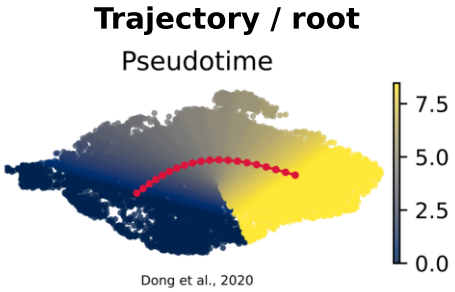
Gast/pSom co-vary (not compete) → single collapsed axis, no bifurcation to test.

STEP FIGURES

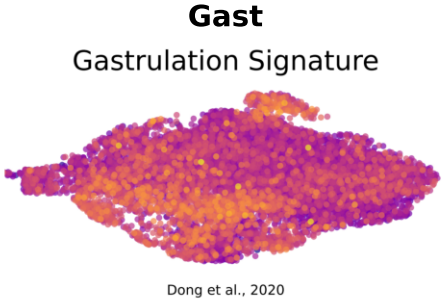
Gate 1: integration



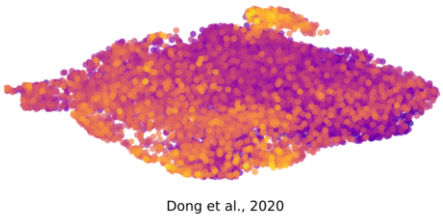
Trajectory / root



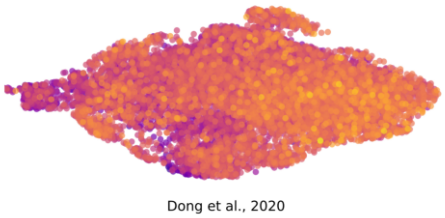
Gast



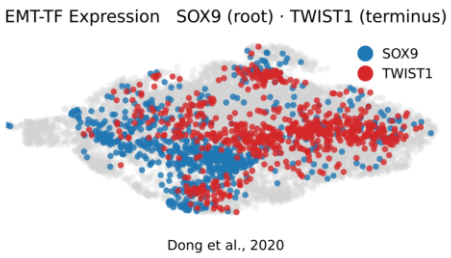
pSom



Crest



EMT-TFs



Dataset. Prostate cancer (Dong et al., 2020); 10x, 4 malignant-bearing patients (5 adeno + 1 small-cell NEPC by clinical histology). 9,698 malignant cells; luminal & neuroendocrine states are poles of one continuous manifold.

Verdict. ✓ Pattern reproduces as a non-branching embryonic axis: Gast/pSom-high origin → Crest-high terminus; developmental order pSom ρ=−0.71, Crest ρ=+0.79 along pseudotime, checked off a diffusion embedding independent of the fitted UMAP. Linear geometry (pSom globally strongest module).

Parameter choices

Min cells/sample	≥100 (no low-count gap; 4/4 kept)	**Integration**	Harmony θ=10; PCA20; spread=1.3
Scope	whole manifold (luminal+NE)	**Trajectory**	scFates curve N=20 λ=0.3 μ=100
Root node	Gast/pSom-high tip	**EMT-TFs**	SOX9 (root) → TWIST1 (terminus)